

Weekly Project Report – Week 3 Unified Generative AI Models via Command-Line Interface

Group 6 Section: 05

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Last Week: DCGAN Implementation (Deep Convolutional Generative Adversarial Network)

- **Objective:** Learn to construct a basic GAN using convolutional layers to generate RGB images.
- **Tasks Completed:**
 - Implemented a **Generator** network using transposed convolution layers (`ConvTranspose2d`) with batch normalization and ReLU activations.
 - Built a **Discriminator** network using convolutional layers (`Conv2d`) with LeakyReLU activations and batch normalization.
 - Applied custom **weight initialization** to ensure stable training.
 - Integrated **BCELoss** for binary classification (real vs. fake) and used the **Adam optimizer** for training both networks.
 - Developed a full **DCGAN training loop**, logging generator and discriminator losses.
 - Generated and visualized synthetic image samples during training using `torchvision.utils`.

Week 3: Conditional GAN (cGAN) Implementation

- **Objective:** Extend GAN architecture to condition image generation on class labels (e.g., MNIST digits).
- **Tasks Completed:**
 - Designed a **Generator** that combines random noise with embedded class labels using `nn.Embedding` and fully connected layers.
 - Constructed a **Discriminator** that takes both the image and its label to verify authenticity and consistency.
 - Implemented label-conditioned training using concatenation of embeddings with image vectors.
 - Built a robust **training loop** with real-time progress tracking via `tqdm`.
 - Enabled **class-conditional sampling**, allowing the model to generate specific digit classes (e.g., only "3" or "7").
 - Added functionality to **save generated samples** as image grids.

Skills Gained:

- Proficiency in PyTorch for advanced model building and training.
- Understanding of GAN fundamentals including adversarial training dynamics.
- Mastery of convolutional and linear layers in generation and discrimination tasks.
- Use of embeddings for label conditioning in neural networks.
- Visualization and debugging of deep learning training processes.